

Oisín Davey

oisindavey02@gmail.com | (+353) 087 390 8166 or (+44) 075 0866 9795 | [LinkedIn](#)

Education

PhD | Theoretical Physics | University of Cambridge | 2026 – 2030.

Professor Matthew Wingate, studying tensor networks and quantum computing methods for the simulation of quantum chromodynamics on the lattice.

MASt | Theoretical Physics | University of Cambridge | 2025 – 2026.

Graduated with Merit from Part III of the Mathematical Tripos. Awarded the Jennings prize from Wolfson College. During the Lent-Easter break I attended the UNIVERSE+ school for Positive Geometry in Les Houches under a research travel grant. I also delivered a seminar to other Part III students on the fractal spectrum of the Harper-Hofstadter model.

BSc | Theoretical Physics & Pure Mathematics | Maynooth University | 2021 – 2025.

Awarded 1.1/First Class Honours. Awarded the Hamilton Prize for the top 9 undergraduate students of mathematics in Ireland along with all 8 other prizes available in my course of study. Ranked 1st amongst all science students. Founded Physics & Chemistry Society.

Academic Projects

MASt | Part III Essay, Dissertation | Chiral Symmetry on the Lattice.

- I studied the emergence of the chiral anomaly of massless fermions in low-dimensional lattice models through the non-local Onsager algebra in the continuum limit.
- Awarded a grade of 82%.
- Available [here](#).

BSc | Final Year Project | Dynamical systems within Anderson Localisation and the Harper-Hofstadter model.

- I investigated the spectral and dynamical properties of two tight-binding Hamiltonians, the first describing Anderson Localisation, and the second being the Harper-Hofstadter model. I studied the appearance of cocycles and Lyapunov exponents in these systems.
- Awarded a grade of 86%, the 2nd-highest in my cohort.
- Available [here](#).

BSc | Third Year Project | Computation and statistical analysis of the first million zeros of Riemann's zeta function vis-à-vis Montgomery's pair correlation conjecture.

- Studied critical line zeros of the Riemann zeta function apropos the connection between their distribution and the Gaussian Unitary Ensemble. Resulted in original proposal of a fast approach for computing complex power sums to allow for fast computation of the Hardy Z function and thence Riemann's zeta function. Project was done in collaboration with another student.
- Awarded a grade of 98%, the highest in our cohort.
- Available [here](#).

Skills

Programming language acquisition: Comfortable and practiced adapting to languages to suit the needs of a project; having incorporated C++ & Python in major projects. Well versed with Git from various projects.

Tutoring: Annually contracted by University College Cork training our international Programming Olympiad team; academic tutor for 1st science at Maynooth University; volunteer training the national Mathematics Olympiad class. My work in education resulted in my membership within the Irish Mathematical Trust.

Data structures & Algorithms: Represented Team Ireland for three consecutive years in the International Programming Olympiad (IOI), winning a bronze medal in Singapore 2021. Placed 1st in the All-Ireland Collegiate Programming contest in 2023. Represented Maynooth University at UKIEPC (University of Edinburgh) and NWERC (TU Delft) in ICPC regionals.

Experience

Mathematics Internship | Maynooth University – Kildare | June 2025 – September 2025

- Review of Besicovitch sets in two dimensions. Showed that the limiting closed Perron tree figure has Hausdorff dimension 2 with null Hausdorff measure. Investigated the Baire category of the space of Besicovitch sets in the projective plane among the space of omnidirectional line sets.

Software Development Internship | CERN - Geneva | June 2024 – September 2024

- Developed the proprietary graphical tool "Vis-à-Gis" using the geodesy api PyQGIS and Qt to aid the beam surveyors in analysing the results of "Logiciel Général de Compensation", detailing the measurement network precision data from surveys of the Large Hadron Collider.
- Created, for Vis-à-Gis, a novel computational method for displaying projections of confidence ellipsoids using spectral approaches.
- Honed my French, the operative language of the section for internal documentation.

Photonics Research Fellowship | Tyndall - Cork | June 2023 – September 2023

- Created original C++/Python model for the spectra of quantum-confined stark effect based electro-absorption modulators.
- Using time complexity analysis, I optimised the efficiency (from cubic to log-linear in the number of lattice sites) of the old model by using the iterated Arnoldi method. I reframed a term in the model as a convolution computable via fast-fourier transforms.

Research & Development Internship | SENSUS - Athlone | June 2022 – August 2022

- Agri-tech startup, worked in soil science and meteorological data science.
- Using QGIS, created personalized data reports/visualisations for farming clients, detailing correlations of rainfall, crop hue (obtained via a satellite imagery API), cloud cover, temperature, etc. Computed statistical correlations with C++.
- Developed proprietary digital soil penetrometer prototype to assess soil bulk density in the field. The device sent stress/strain and real-time kinematics data for a standard soil probe via radio to a GNSS base station. The GNSS computations required me to work with RTKLib, embedded systems, and Python.

Referees

Academic Referees

- Dr. Graham Kells, Associate Professor of Physics at Maynooth University. Supervised my Final Year Project and taught me Statistical Mechanics, Thermal Physics, and Quantum Mechanics II. Contact: graham.kells@mu.ie
- Dr. Paul Watts, Lecturer of Physics at Maynooth University. Taught me General Relativity, Classical Mechanics, and Fluid Mechanics. Contact: paul.watts@mu.ie
- Dr. Jon-Ivar Skullerud, Associate Professor of Physics at Maynooth University. Supervised my Computational Physics Project and taught me Waves and Vibrations, Quantum Mechanics I. Contact: jonivar.skullerud@mu.ie

Professional Referees

- Prof. Frank Peters, Professor of Physics at University College Cork. Supervised my summer fellowship at Tyndall National Institute. Contact: f.peters@ucc.ie
- Guillaume Kautzmann, Survey Engineer in the Beams department at CERN. Supervised my work during my summer placement in CERN. Contact: guillaume.kautzmann@cern.ch